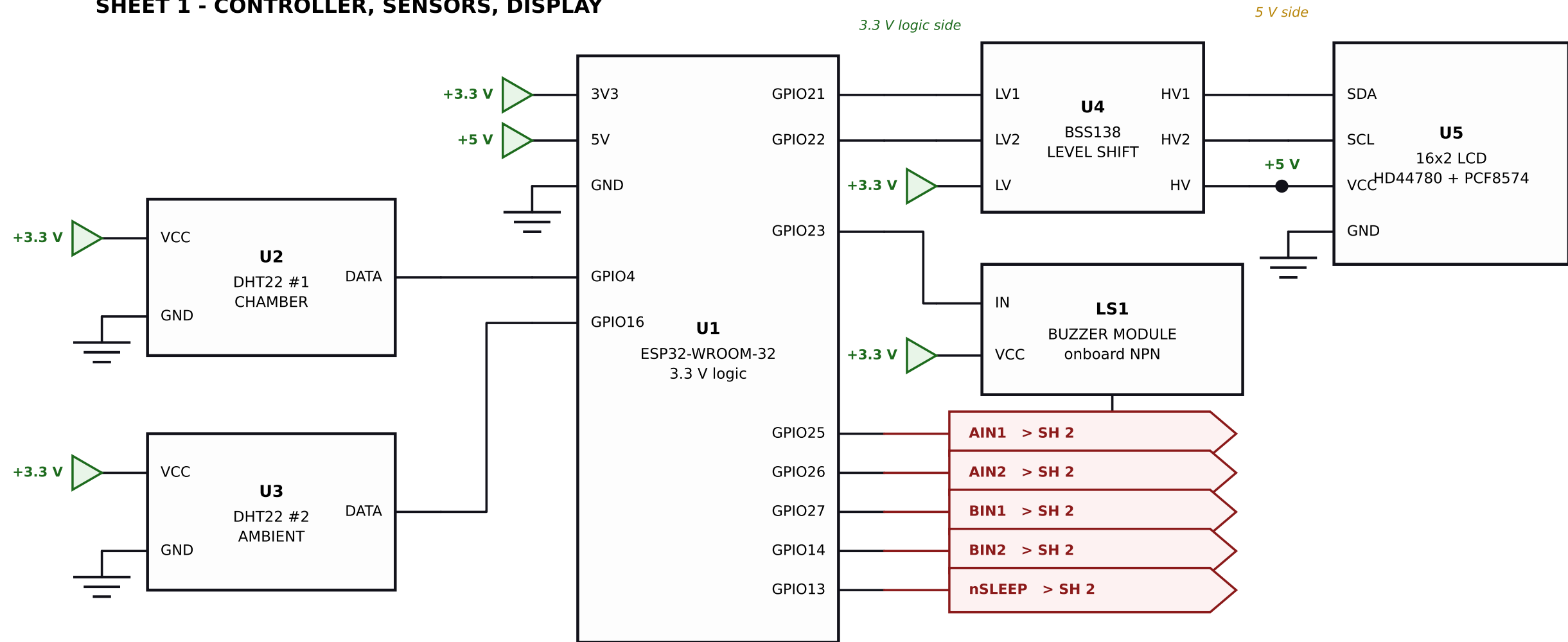


SHEET 1 - CONTROLLER, SENSORS, DISPLAY



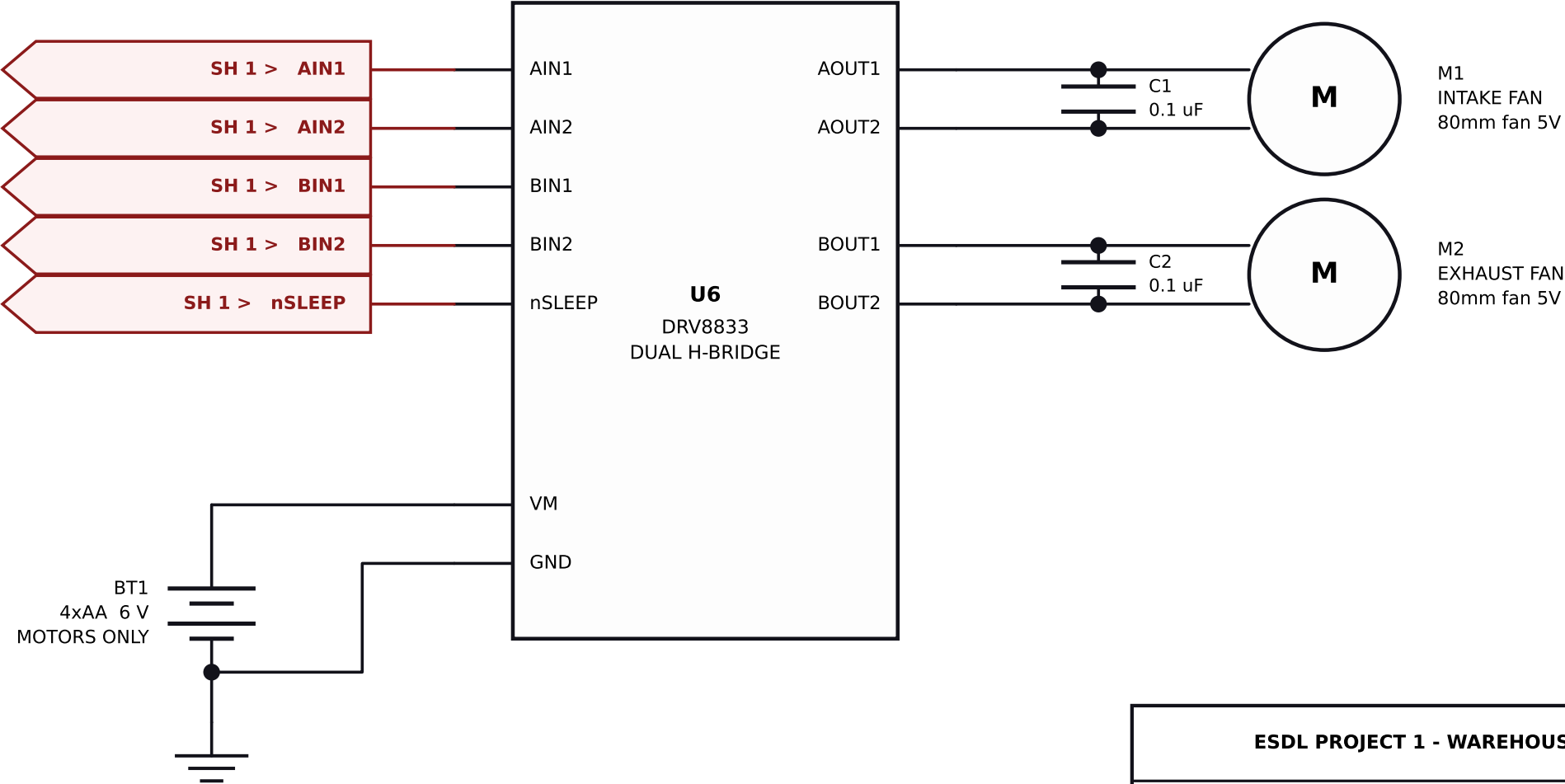
All grounds common. DHT22 data lines are 3.3 V logic and connect directly to the MCU.
The LCD is the only 5 V device: its I2C lines MUST pass through U4.
The +5 V net originates at the ESP32 5 V pin (USB-derived).

ESDL PROJECT 1 - WAREHOUSE VENTILATION		
Controller, Sensors, Display		
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SHEET 2 - MOTOR DRIVE AND POWER

COMMON GROUND IS MANDATORY

BT1 negative, U6 GND and the ESP32 GND must all tie together, or the H-bridge sees no valid logic level and the motors will not run.



C1 and C2 are soldered AT THE MOTOR TERMINALS, not on the breadboard - brush noise on the I2C bus is the most common failure in this build. nSLEEP must be driven HIGH or the bridge stays off. Motor supply is separate from logic.

ESDL PROJECT 1 - WAREHOUSE VENTILATION		
Motor Drive and Power		
TJ / Tabitha	2026-09-16	REV A SH 2/2